SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.
(5 Marks)

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.
(5 Marks)

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. **(5 Marks)**

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.
(5 Marks)

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency.
(5 Marks)

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance**.** **(5 Marks)**

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

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1. Ride-Sharing System (Driver–Passenger Matching)

Objective

Minimize passenger waiting time and improve service quality.

Why RL?

The environment is dynamic (traffic, demand fluctuations), so RL can learn optimal matching over time.

MDP Components

- **States:** Driver location, passenger request, traffic, driver availability
- **Actions:** Assign driver, reject request, reassign another driver
- **Environment:** Road network with real-time conditions

Reward Function

- +10 for quick pickup
- −5 for long waiting time
- −3 for inefficient routing
- +8 for high customer rating

Policy

Maps state → best driver selection decision

Learning Process

- Uses Q-learning/Deep RL
- Learns from past assignments and outcomes

Continuous Learning

Improves matching efficiency as more data is collected

Flow (RL Architecture)

State → Action → Environment → Reward → Policy Update → Repeat

Expected Outcomes

- Reduced waiting time
- Better driver utilization
- Higher customer satisfaction

2. E-Learning Platform (Personalized Recommendation)

Objective

Provide personalized study material effectively

Why RL?

Learner preferences change → RL adapts continuously

Exploration vs Exploitation

- **Exploration:** Suggest new topics/content
- **Exploitation:** Recommend familiar/preferred content

States

User progress, performance, interests

Actions

Recommend lesson, quiz, or topic

Reward Function

- +10 for course completion
- +5 for engagement
- −3 for skipped content

Impact

Balanced strategy:

- Too much exploitation → limited learning
- Too much exploration → confusion

Outcome

Better learning outcomes and engagement

3. Robotic Vacuum Cleaner

Objective

Clean efficiently while avoiding obstacles

MDP Components

- **States:** Room layout, dirt level, obstacles
- **Actions:** Move, turn, clean, stop

Policy

Defines best movement strategy in each state

Reward Function

- +5 for cleaning dirt
- −2 for collision
- −1 for unnecessary movement

Value Function

Estimates long-term reward of a state

Learning Process

- Learns optimal cleaning path
- Improves through repeated trials

Continuous Improvement

Better navigation and efficiency over time

Outcome

- Faster cleaning
- Reduced collisions
- Energy efficiency

4. Cryptocurrency Trading System

Objective

Maximize profit through trading decisions

Actions

Buy, sell, hold

States

Market trends, price history, indicators

Reward Function Challenges

- Profit is delayed
- Market volatility
- Risk management

Poor Reward Design Issues

- Short-term rewards → risky trades
- Ignoring risk → large losses
- Overfitting to past trends

Example

Reward only profit → agent may take extreme risks

Solution

Balanced reward:

- Profit − risk penalty

Outcome

More stable and reliable trading strategy

5. Smart Irrigation System

Objective

Optimize water usage and crop health

MDP Components

- **States:** Soil moisture, weather, crop stage
- **Actions:** Increase/decrease irrigation

Reward Function

- +10 for optimal crop growth
- −5 for water wastage
- −3 for under-irrigation

Policy

Determines optimal watering schedule

Learning Process

- Uses environmental feedback
- Adjusts irrigation strategy

Continuous Learning

Adapts to weather and seasonal changes

Outcome

- Water conservation
- Improved crop yield
- Sustainable farming

Code of Conduct:

*I **E.V.Mahendra Reddy(192425214)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mahendra reddy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution

Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand